

Huffman Coding

Algorithm:

- ① Write characters & frequency in tabular form.
- ② Now write in ascending order
- ③ Add first two least frequency, find their sum is in root node and first no is left child second one is right child.
- ④ Check if there are in ascending order if some frequency.
 - (i) Character vs Character $\rightarrow$ Alphabetical order
 - (ii) Character vs Tree $\rightarrow$ Character first
 - (iii) Tree vs Tree $\rightarrow$ earlier formed tree first.
- ⑤ Repeat this process until you get one final tree.
- ⑥ After this left child is 0 right child is 1, do it for full tree.
- ⑦ Write each letter codes (in 0 or 1)
- ⑧ Then find compressed bits using formulae.

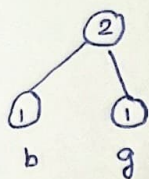
$$\text{compressed bits} = \sum \text{code length} \times \text{frequency}$$

Sol:

Character	d	a	t	n	l	y	i	c	s	e	g	b	o	r
Frequency	2	7	4	4	4	4	2	3	2	1	3	1	2	2

①

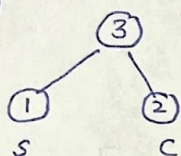
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b q s c d a r y e i l n t a



1 2 2 2 2 2 3 3 4 4 4 7
s c d o r y e i l n t a

1 2 2 2 2 2 2 3 3 4 4 4 7
s c d o r y b g e i l n t a

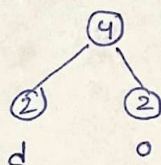
②



2 2 2 2 2 3 3 4 4 4 7
d o r y b g e i l n t a

2 2 2 2 2 3 3 4 4 4 7
d o r y b g e i s c l n t a

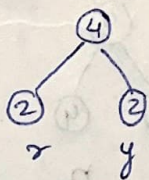
③



2 2 2 3 3 4 4 4 7
r y b g e i s c l n t a

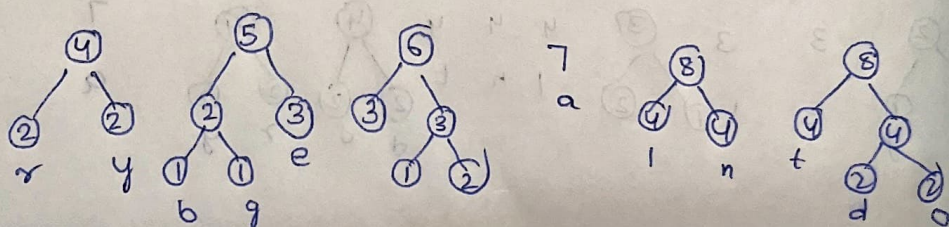
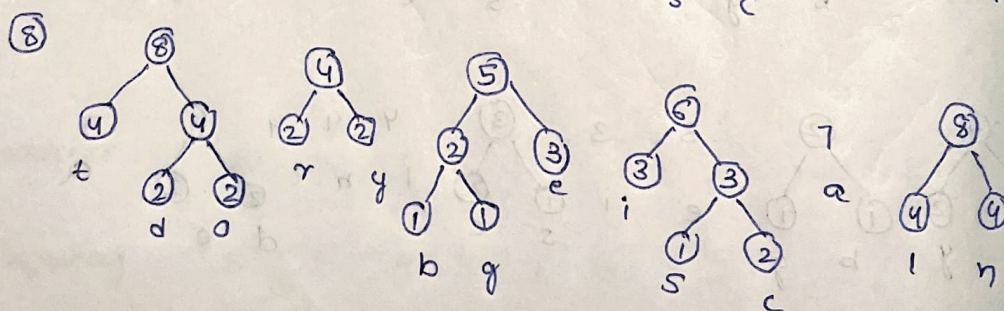
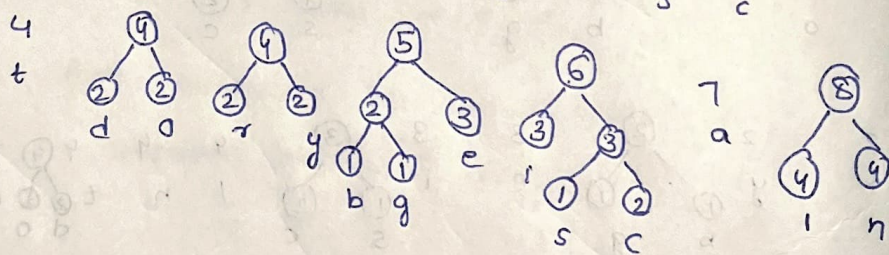
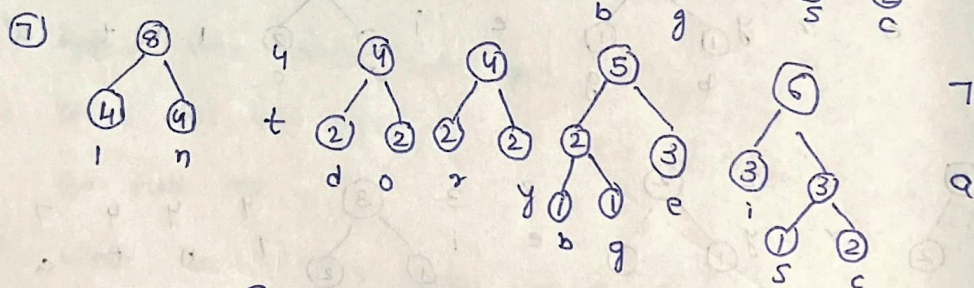
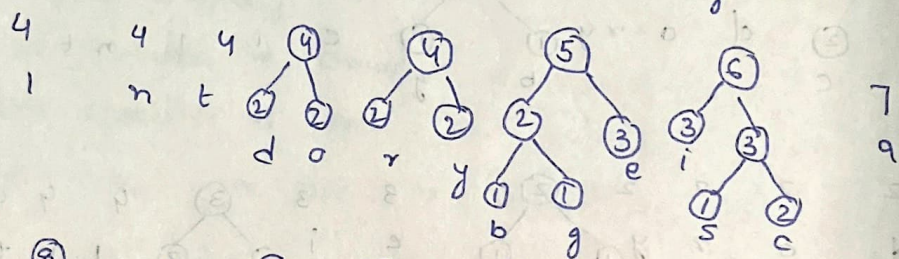
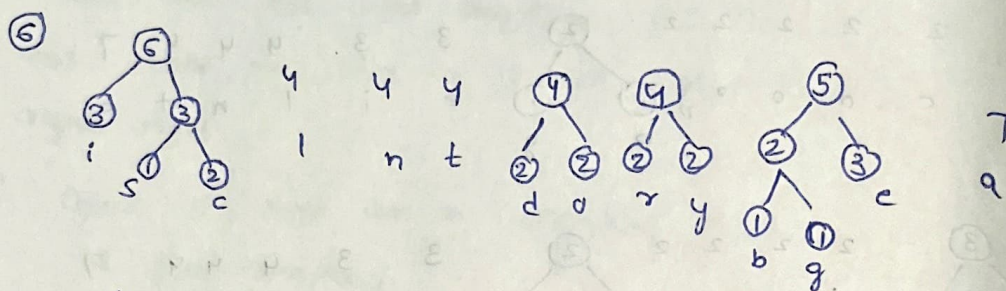
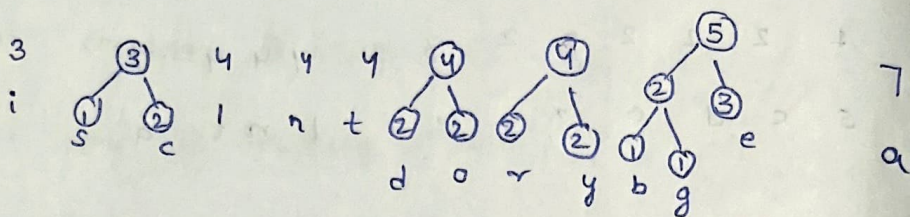
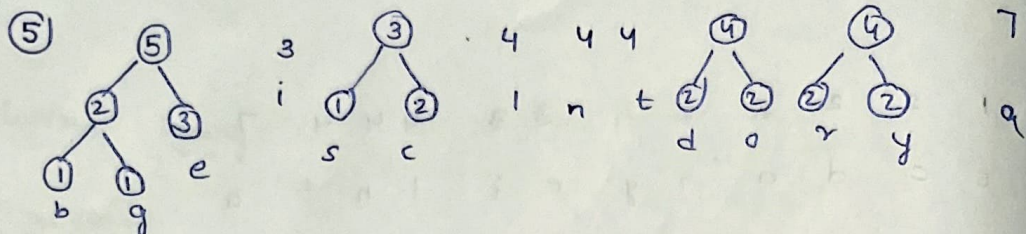
2 2 2 3 3 4 4 4 7
r y b g e i s c l n t a d o

④

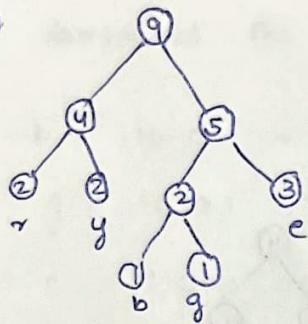


2 3 3 4 4 4 7
b g e i s c l n t a d o

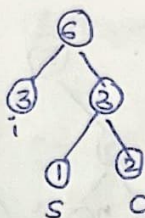
2 3 3 4 4 4 7
b g e i s c l n t a d o r y



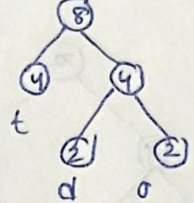
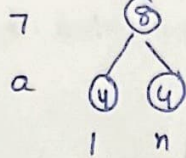
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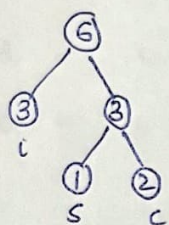
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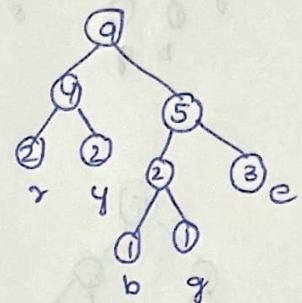
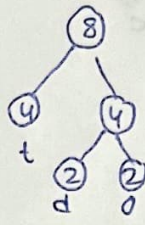


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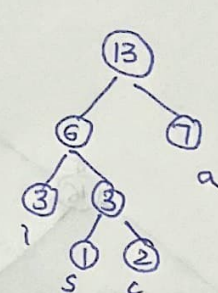
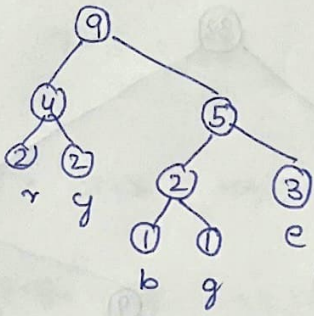
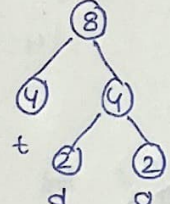
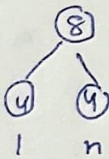
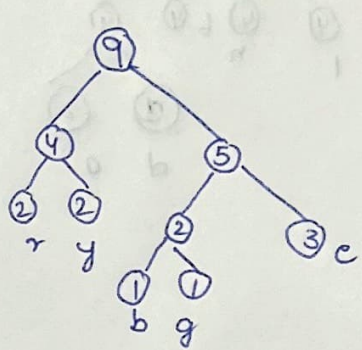
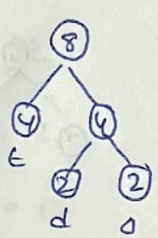
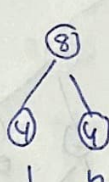
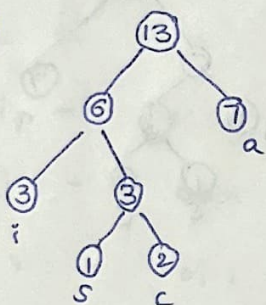


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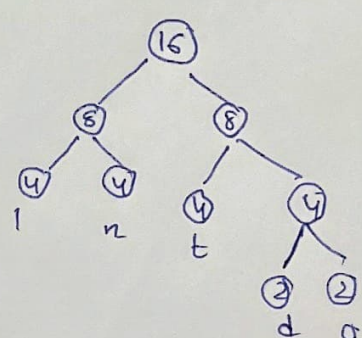
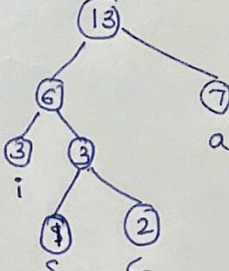
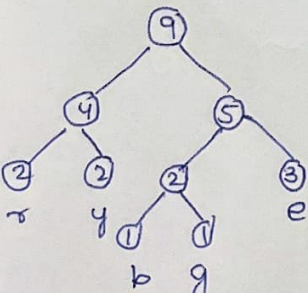
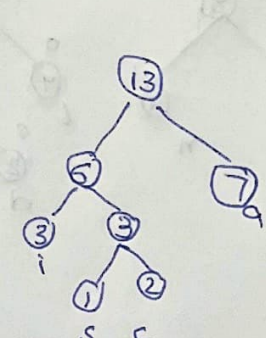
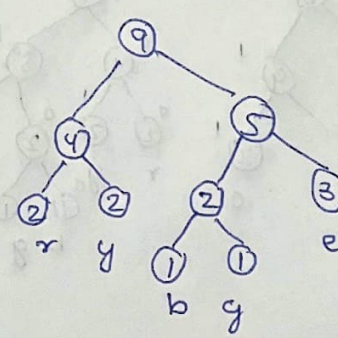
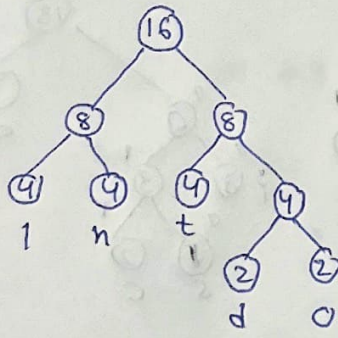
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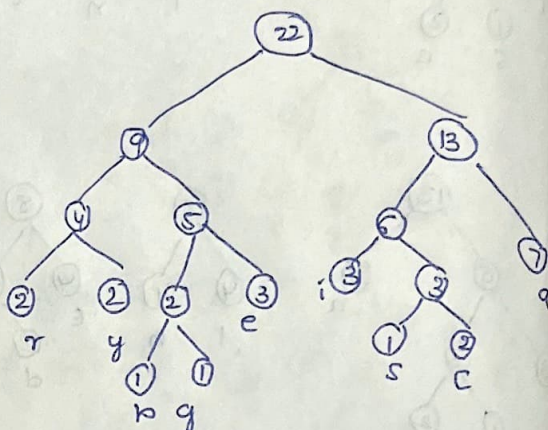
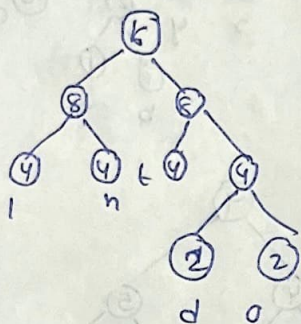
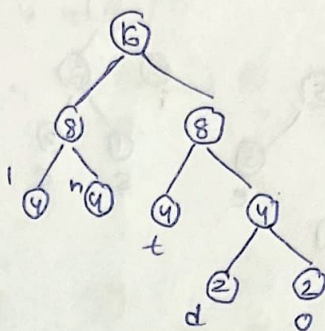
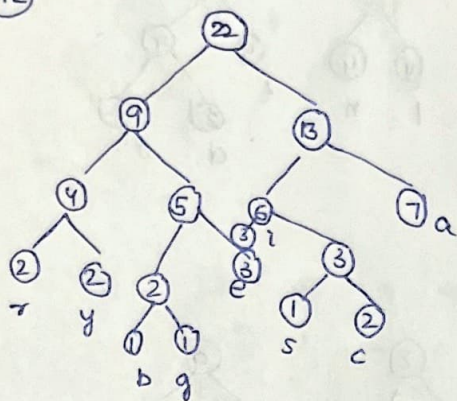
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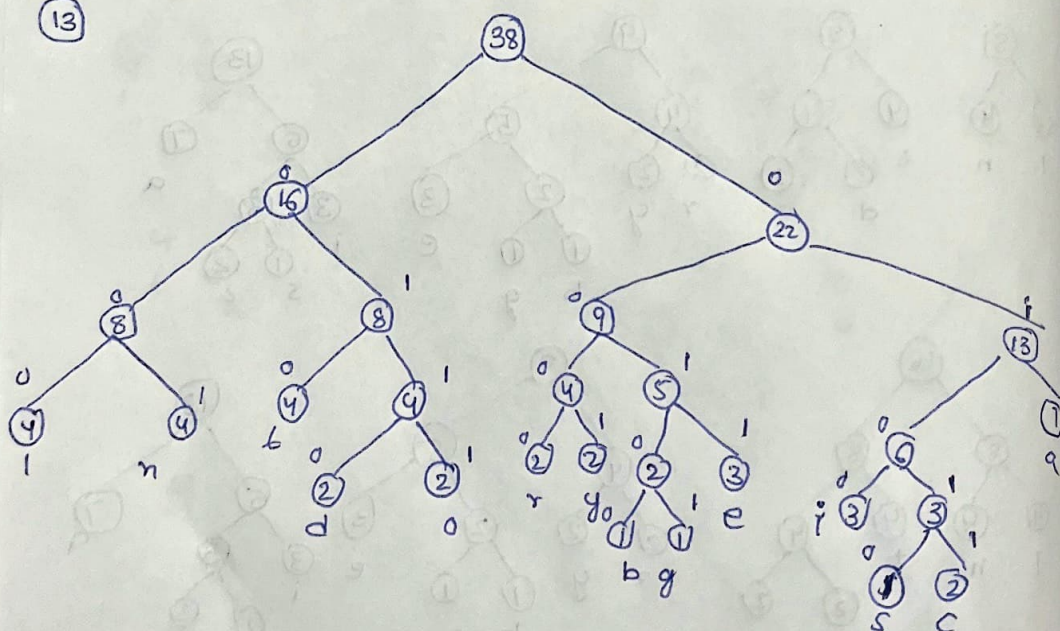
11



12



13



$$\text{Compressed Bits} = \sum \text{frequency} \times \text{code length}$$

$$b - 10100 = 1 \times 5 = 5$$

$$g - 10101 = 1 \times 5 = 5$$

$$s - 11010 = 1 \times 5 = 5$$

$$c - 11011 = 2 \times 5 = 10$$

$$d - 0110 = 2 \times 4 = 8$$

$$o - 0111 = 2 \times 4 = 8$$

$$r - 1000 = 2 \times 4 = 8$$

$$y - 1001 = 2 \times 4 = 8$$

$$e - 1011 = 3 \times 4 = 12$$

$$i - 1100 = 3 \times 4 = 12$$

$$l - 000 = 4 \times 3 = 12$$

$$n - 001 = 4 \times 3 = 12$$

$$t - 010 = 4 \times 3 = 12$$

$$a - 111 = 7 \times 3 = 21$$

$$\text{Original bit length} = 38 \times 8 = 304 \text{ bits}$$

$$\begin{aligned} \text{Compressed bits} &= 5 + 5 + 5 + 10 + 8 + 8 + 8 + 8 + 12 + 12 + 12 + 12 + 12 + 12 + 21 \\ &= 138 \text{ bits} \end{aligned}$$

$$\begin{aligned} \text{Saved bits} &= 304 - 138 \\ &= 166 \text{ Bits.} \end{aligned}$$